Design of a Temperature-Compensated PLL Frequency Synthesiser

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Guide's Signature with Date

Signature: _____

Date: _____

Problem & Objectives

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Problem

A ring-VCO drifts with temperature: threshold voltage, mobility and capacitance all shift, so the same control voltage gives a different frequency hot vs cold. Process corners change the size of the drift.

Objectives

Integer-N charge-pump PLL in sky130B, open-source and headless (ngspice 47): REF 28.275 MHz → PFD → charge pump → loop filter → 5-stage current-starved ring VCO → /8.

Measure the uncompensated drift on every process corner, design a temperature-compensation, and validate it with Monte Carlo and hypothesis tests.

Close the loop and check lock time across corners and temperature.

226.2 MHz
target at V_ctrl = 0.9 V (tt, 27 C) = 8 x 28.275 MHz

+/-0.3 %
frequency-drift target over -40 to 85 C

< 2 us
lock-time target

Takeaway: we need a drift figure measured on the real topology and process, then a correction that is proven, not assumed.

Detailed Literature Review

No.	Author(s) & Year	Title / Source	Method / Approach	Findings & Research Gap
1	G. Souliotis et al., 2022	Temperature compensated ring oscillator based VCO, AEU Int. J. Electron. Commun.	PTAT current compensation of a ring VCO inside a PLL; 180nm, 480MHz, fabricated	Closest topology benchmark: -1.5%/+0.4% over -40→120C, measured. First-order only; different node
2	C. Stancu et al., 2023	Temp. and power-supply compensated CMOS clock circuit based on ring osc., Electronics	Temp + supply compensated 50MHz ring-oscillator clock; simulation only	1.6% temperature error pre-layout. A clock circuit, never closed into a PLL
3	Z. Geng et al., 2018	Low power fast locking PLL freq. synthesizer with temp. compensation, EAME	Freq. presetting + dynamic loop BW + static/dynamic temp. comp.; 180nm, 2.1-2.6GHz	Lock <2us over 0-80C, -117dBc/Hz@1MHz, 9mW. Stability claim is the locked output, set by the crystal
4	Y. You et al., 2014	12GHz 67% tuning, 0.37ps RJrms PLL, LC-VCO temp. comp., 0.13um CMOS, IEEE RFIC	LC-VCO PLL for serial links; comp. scheme preserves tuning range; 5.6-13.4GHz	Stays locked -40→85C. Needs inductor models our process supports poorly; not portable to a ring VCO
5	S. Sinha, 2002	Design of an integrated CMOS PLL frequency synthesizer, IEEE MELECON	Dual-loop CP-PLL, 0.35um, 2.4-2.5GHz, 1MHz steps	-82dBc/Hz@10kHz. Architectural reference for our loop; no temperature compensation
6	E. M. V. Freitas et al., 2025	Using open source EDA tools in ASICs for HEP: a mixed comparison, JINST	Open-source vs commercial flows on IHP 130nm blocks incl. a current-starved ring VCO	Digital QoR quantified (+53% area, +118% power); analog VCO comparison qualitative, open-loop, no MC
7	M. Thiriloganathan et al., 2026	2.4GHz LC-VCO fractional-N PLL, open-source, 130nm BiCMOS, SMACD	Fully open-source flow (Xschem/ngspice/KLayout/OpenEMS), EM-modelled inductor	-100.8dBc/Hz@1MHz. Closes a loop open-source, but no temp. compensation, no PVT/Monte Carlo
8	R. N. S. Raphael et al., 2019	PLL-based frequency synthesizer for digital systems driving, BTSym'18 (Springer)	3rd-order CP-PLL for clock generation, integer-N (N=8-16), VCO input-cap-aware LF model	Settling <2us. Same loop class as ours; no temperature compensation

Detailed Literature Review (contd.)

No.	Author(s) & Year	Title / Source	Method / Approach	Findings & Research Gap
9	Y. Li, K. Du, J. Zhang, 2022	Design of ring oscillator with temperature compensation effect, Int. J. RF Microw. CAE	Positive-TC supply voltage on a 31-stage ring, 0.18um; 33kHz sleep timer	120 ppm/C (1.6%) over -40→80C. Single linear coefficient; kHz timer, not a PLL
10	H. H. Hammam et al., 2022	An ultra-low-power process-and-temperature compensated ring oscillator, IEEE ICEEE	3-stage current-starved ring, 130nm, process + temperature compensation	Up to 90% less variation vs. plain current-starved ring, 346nW. Standalone oscillator
11	S. Zhang et al., 2016	Temperature compensation technique for ring oscillators with tail current, Electron. Lett.	Tail current = constant source + PTAT current, first-order	Foundational first-order technique; no numbers reported in abstract; not closed-loop
12	M. Ortner, M. Punzenberger, 2017	Oscillators using quadratic temperature compensation, US Pat. 9,748,962 B2	Linear + quadratic temperature term for a VCO, -40→85C	Generic 2nd-order compensation is patented -- narrows our claim to ring-VCO + closed PLL + open PDK + stats
13	D. Patrinos, G. Souliotis, 2025	Low-voltage, low-power 2.5GHz ring oscillator with process & temp. comp., JLPEA	PTAT current + process-corner detection/tuning; 22nm, 0.8V; post-layout sim	Drift 23.9%→5.4% (tt) and 47.3%→7.2% (PVT). Corner-aware like our C(corner); not in a PLL
14	M. Sivasakthi, P. Radhika, 2024	PVT tolerant hybrid current starved ring VCO, bulk-driven keeper, 45nm, AEU	Two current-starved ring VCOs; Cadence GPDK45; corner + Monte Carlo	6.97GHz, -121.5dBc/Hz@1MHz. PVT tolerance of the VCO itself, no compensation network or loop
15	S. Madheswaran, R. Panneerselvam, 2024	7-stage MCML power-gated current starved VCO for PLL, IJECE	MOS current-mode logic delay cells with power gating, 45nm, simulation	3.06GHz, -125.9dBc/Hz. Same device class as ours; no closed-loop temperature characterisation

- Research gap (from all 15): no reported work combines a compensated ring VCO inside a closed loop, characterised across process corners with Monte Carlo rigor -- that gap is what this project fills.

Novelty, Patent and Research Gap

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What is NOT novel

Generic quadratic (2nd-order) temperature compensation of an oscillator is patented: US 9,748,962 B2 (Ortner and Punzenberger, Infineon, 2017) [12]. We do not claim it.

Current-starved ring VCOs and the charge-pump PLL topology are textbook.

What this project adds

A closed-loop, per-corner, statistically validated study of a compensated ring-VCO PLL on an open PDK (sky130B), with the failures and corrections documented.

Research gap (from 15 IEEE-level references)

No reported work combines a compensated ring VCO inside a closed loop, characterised across process corners with Monte Carlo rigour.

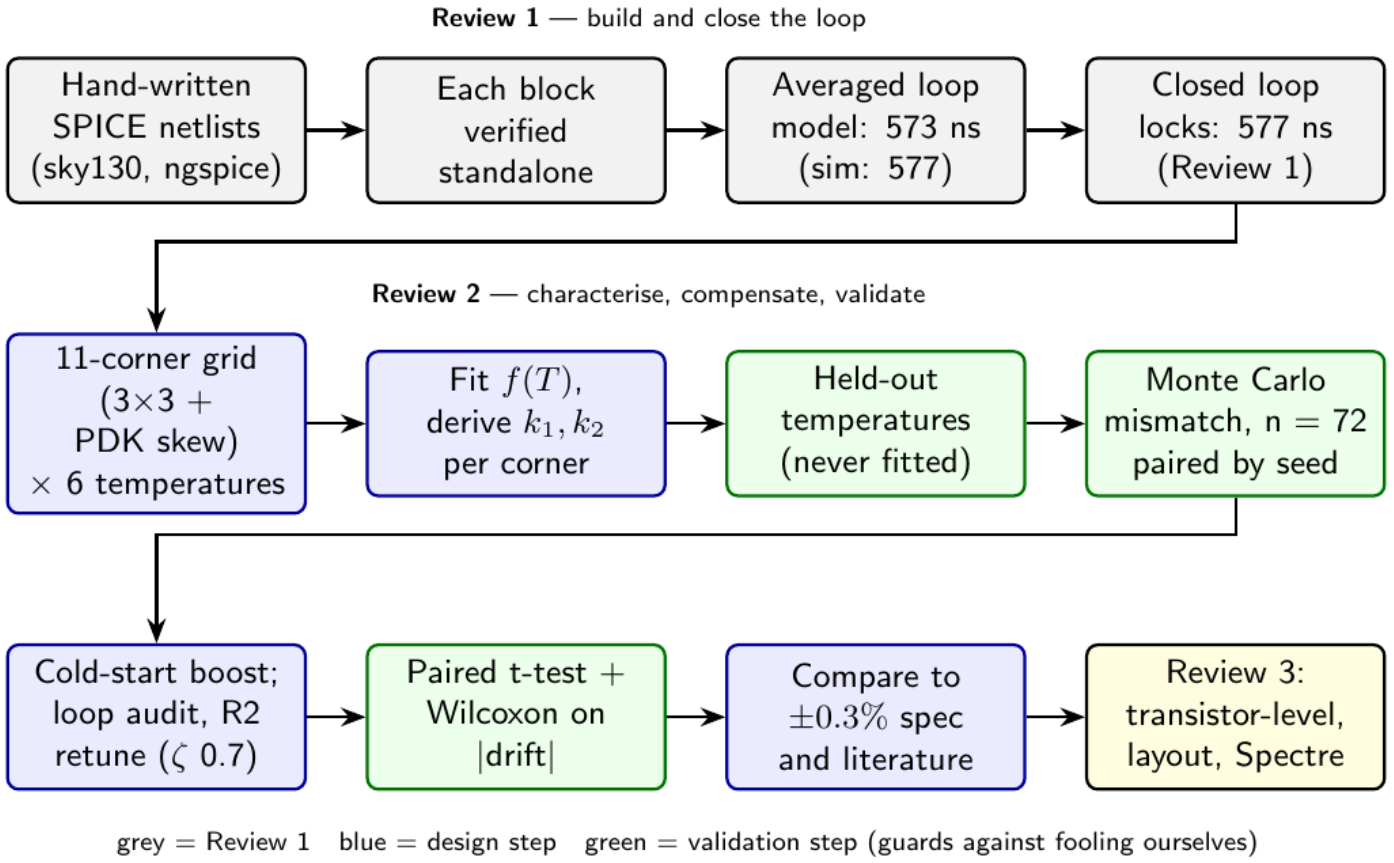
15 refs
read; closest: Souliotis 2022 [1], Stancu 2023 [2], Geng 2018 [3]

1 patent
narrowed the claim to ring-VCO + closed PLL + open PDK + statistics

Open PDK
everything reproducible: ngspice + sky130B, no licensed tools yet

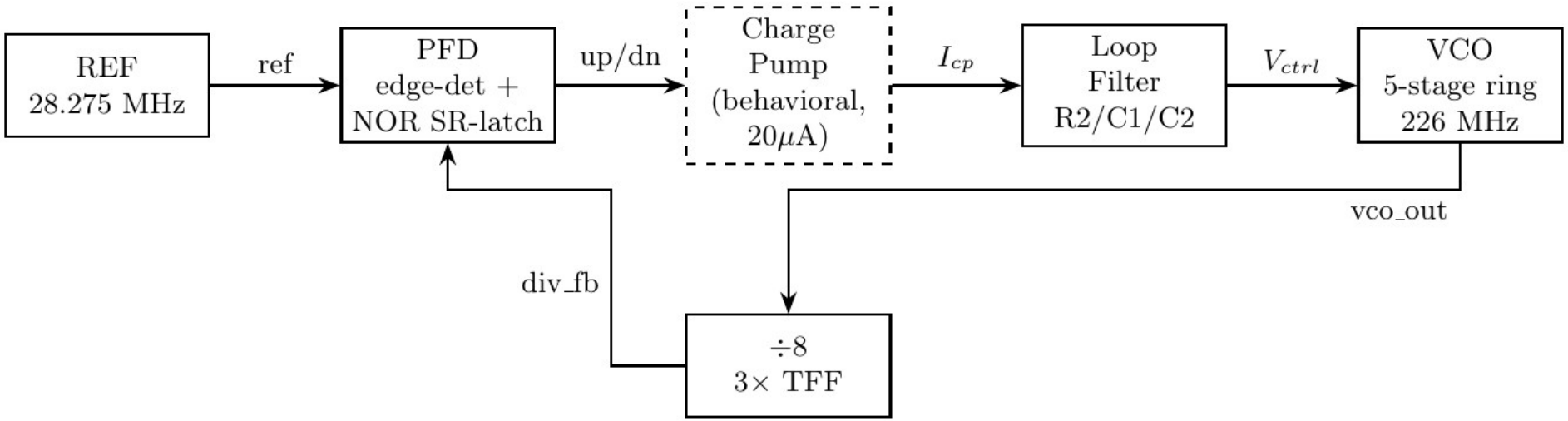
Takeaway: our contribution is the validated closed-loop study, not the idea of 2nd-order compensation.

Methodology: Review 1 → Review 2 → Review 3



Takeaway: characterise, compensate, close the loop, then validate statistically.

System Architecture



Closed-loop lock (tt, 27°C): **503 ns** (R2 = 3.6 kΩ; spec <2 μs)

20 uA
charge pump (behavioural), 5x until first divider edge

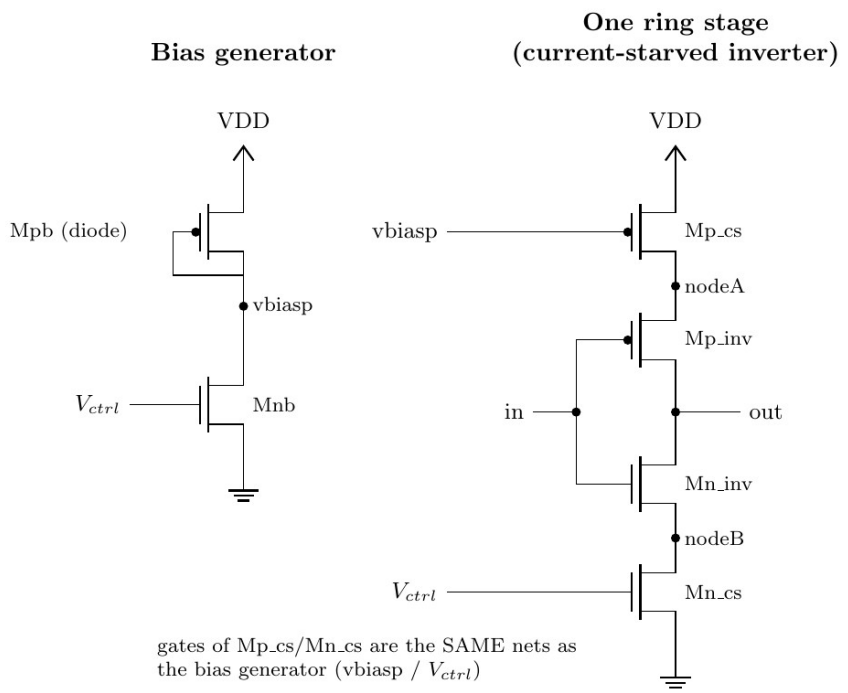
3.6 kOhm
R2 (retuned from 1.7 k); C1 68 pF, C2 6.8 pF

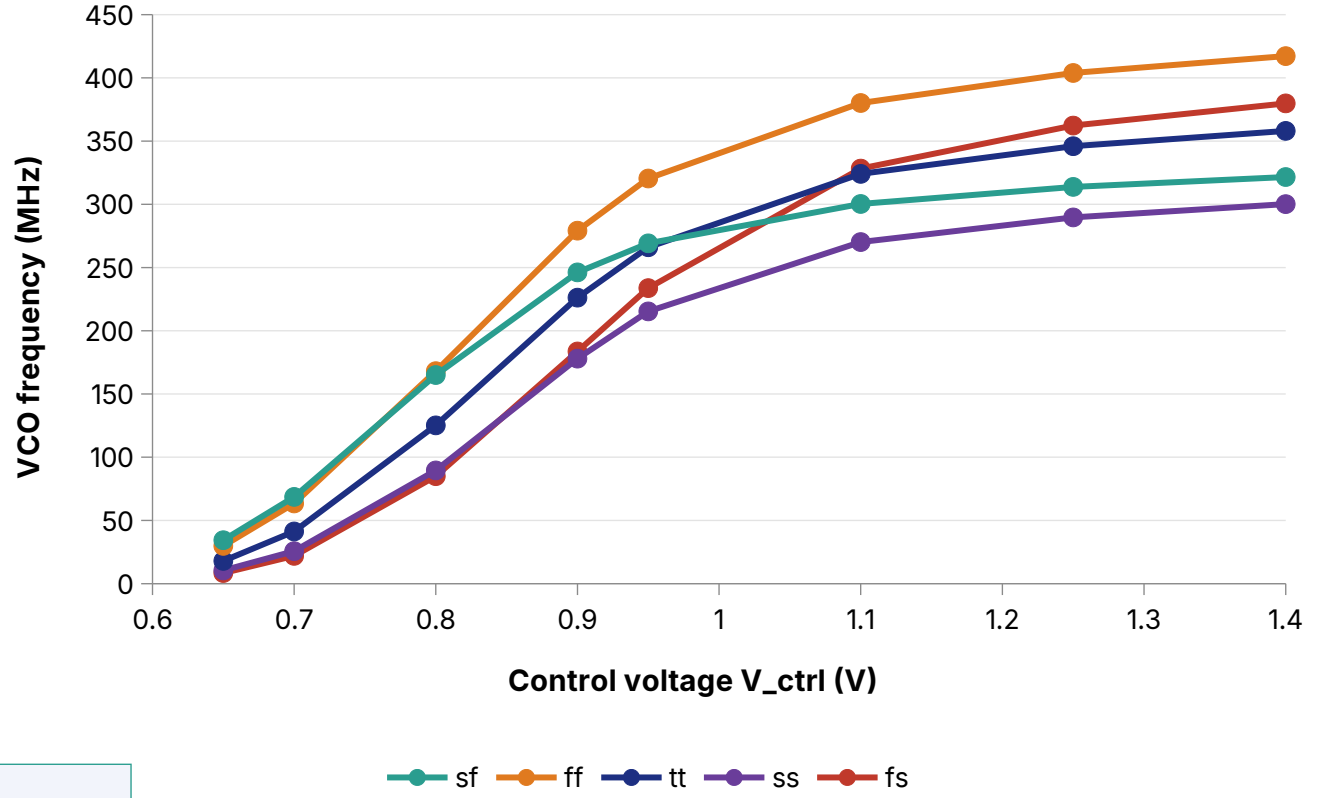
5 stages
current-starved ring VCO, tunes 18 → 358 MHz

/ 8
divider (3 T flip-flops)

Takeaway: a standard integer-N loop, so every difference we see comes from the VCO and its compensation.

VCO: Schematic & Tuning Curve



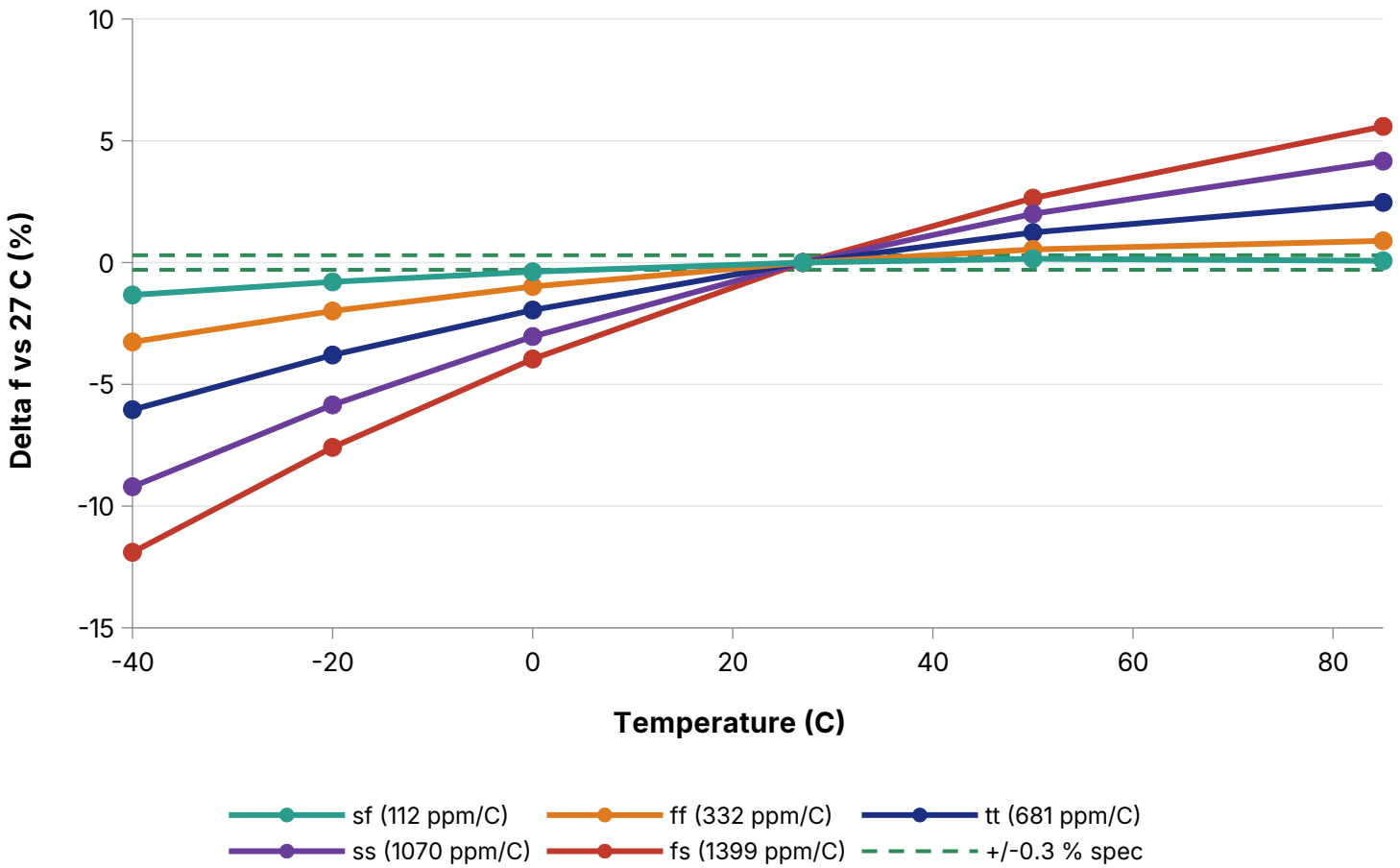


~890 MHz/V
local K_VCO at 0.9 V (tt)

18 → 358 MHz
over 0.65 → 1.40 V (tt)

Takeaway: V_ctrl sets the starving current; 0.9 V gives the 226 MHz target. K_VCO is ~890 MHz/V (an earlier 669 was wrong).

Uncompensated Drift: PDK Corners



12.5x
tempco spread, sf 112 to fs 1399 ppm/C

8.51 %
tt frequency span, -40 to 85 C

112 - 1399
ppm/C across the 5 PDK corners

Takeaway: every corner is far outside +/-0.3 % without compensation, and the corners disagree by a factor of 12.

Why the Corner Numbers Looked Out of Range

In review, ff / tt / fs / sf values were not in the ranges they are meant to be. We audited it (ngspice 47, same PDK).

Verdict: the simulations were fine. Three things were wrong in how we expected and described the corners.

1. Labels

In this PDK sf = fast-NMOS / slow-PMOS and fs = slow-NMOS / fast-PMOS (measured). Earlier labels were reversed.

2. Expectation

'Skew corners lie between ss and ff' is false in sky130: its skew corners shift each device further than ss/ff do. At the VCO's low bias this is amplified to about 9 %.

3. Tempco is not bounded by ff and ss

Even in a clean textbook NMOS x PMOS set the tempco extremes sit at the skew corners.

Takeaway: nothing was broken; our expectation of the corner envelope was wrong.

Audit 1: PDK Skew Corners Are Wider Than ss / ff

Threshold shift vs tt (mV)	ss	ff	PDK sf	PDK fs
NMOS Vth	+25	-25	-36	+36
PMOS |Vth|	+15	-23	+32	-43

The extracted device thresholds of the skew corners move further than ss/ff.

At the VCO bias (0.9 V, overdrive only ~0.2 V) an extra ~11 mV of Vth is ~9 % of drive current.

At 1.8 V the same corners agree within 1 %.

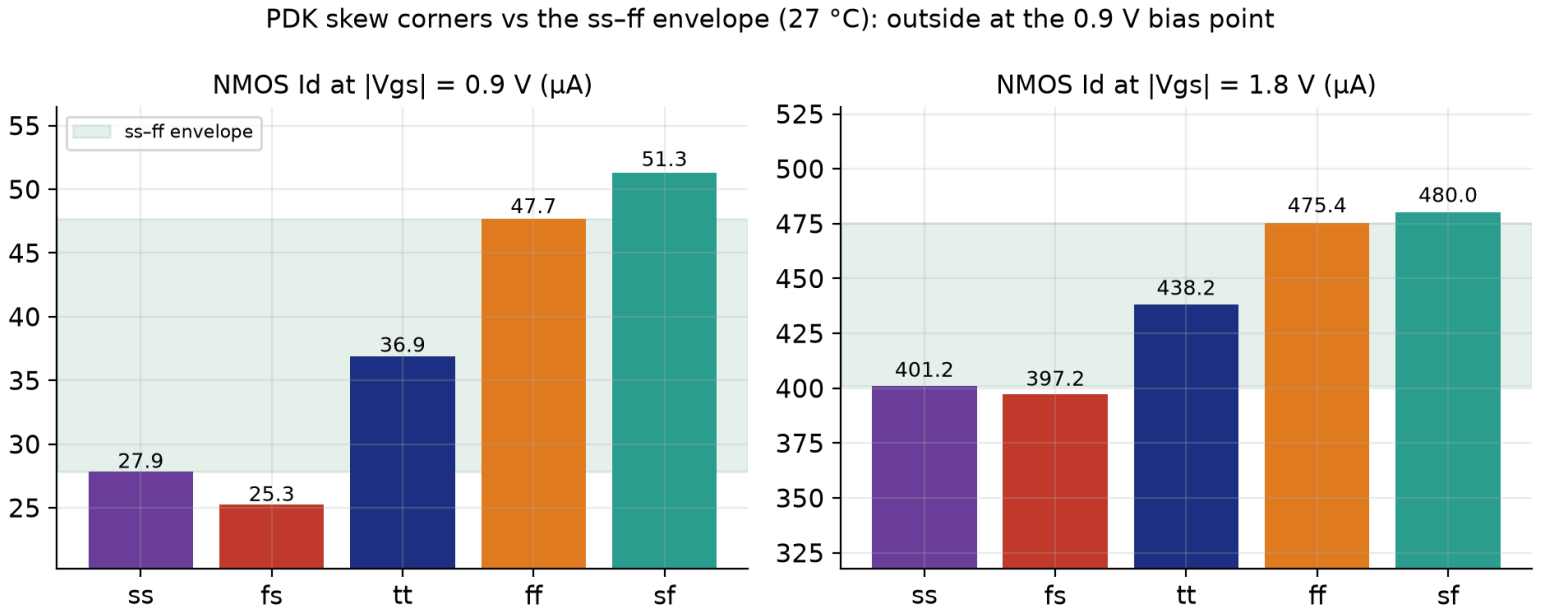
~9 %
skew corners fall outside the ss-ff envelope at 0.9 V

< 1 %
outside the envelope at 1.8 V

11 corners
9 textbook + PDK sf + PDK fs

Takeaway: the PDK defines its skew corners more aggressively than ss/ff, so they can fall outside them.

Audit 2: Device Current vs the ss-ff Envelope



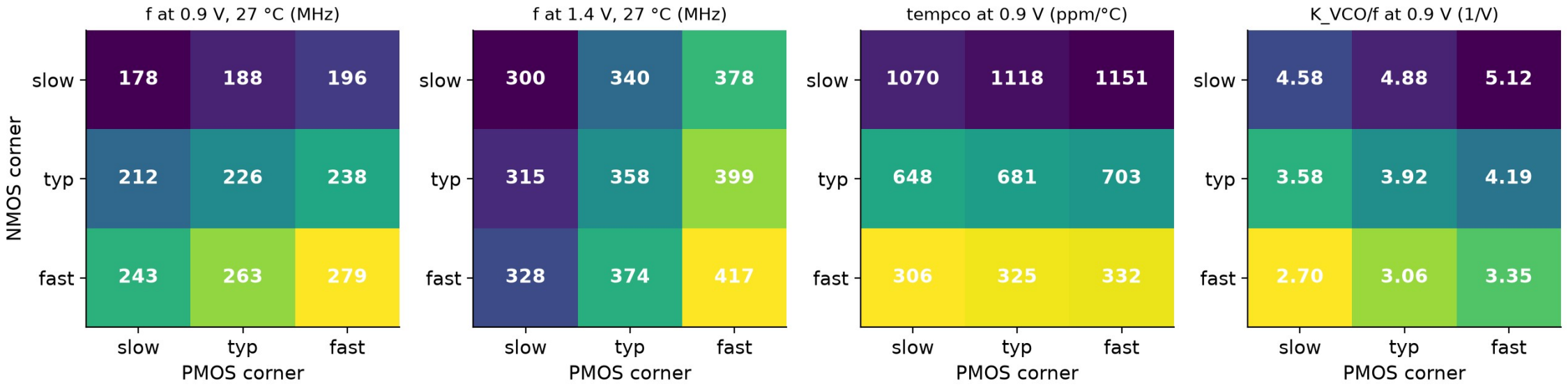
25.3 - 51.3 uA
NMOS Id at 0.9 V: fs 25.3, ss 27.9, tt 36.9, ff 47.7, sf 51.3

397 - 480 uA
NMOS Id at 1.8 V: skews within 1 % of ss/ff

8 - 9 %
skews outside the ss-ff envelope at 0.9 V

Takeaway: at the low VCO bias, skew corners step outside ss/ff; at full gate drive they do not.

Audit 3: A Textbook 3 x 3 Corner Set



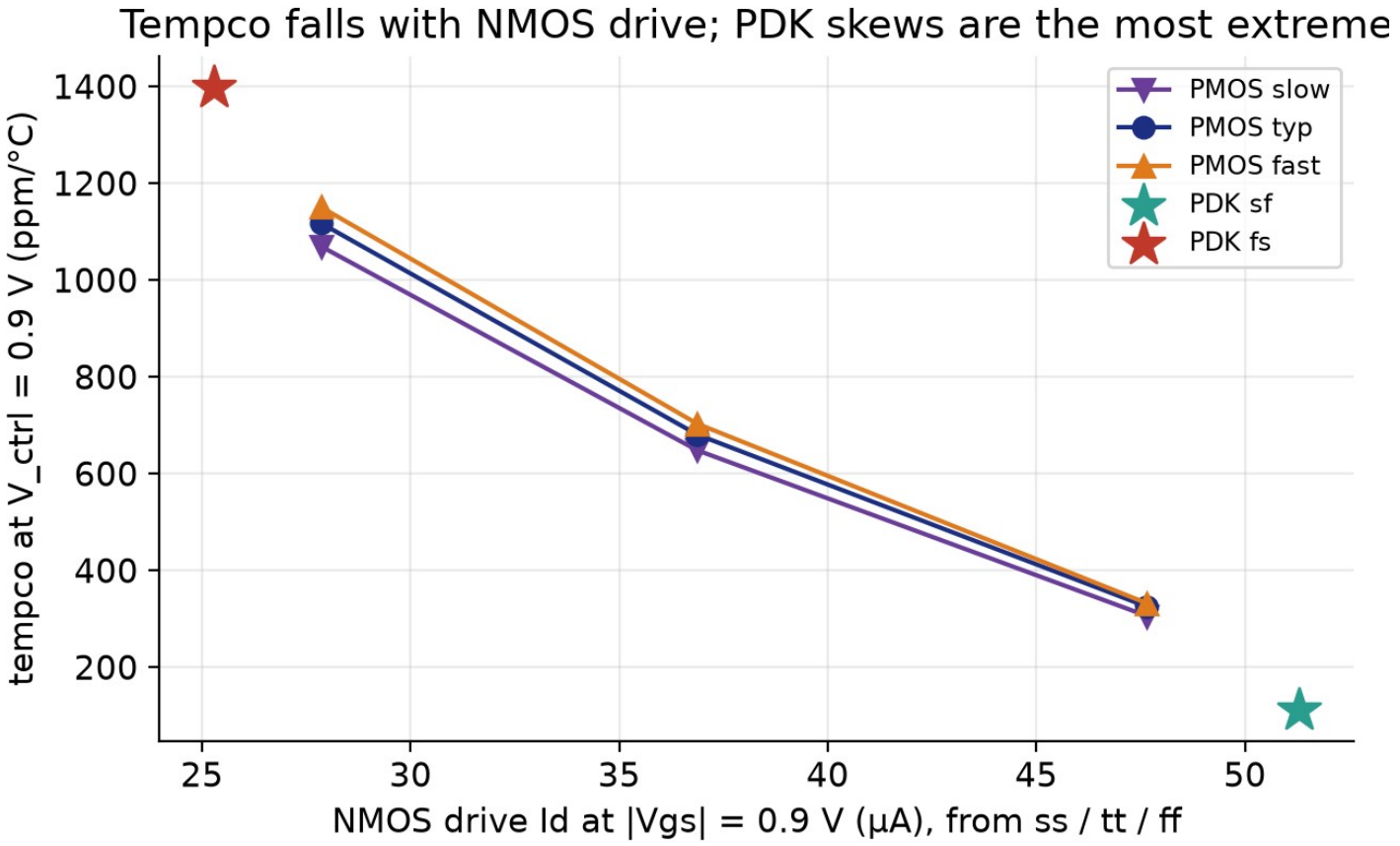
8 / 8
ordering checks pass (f rises with N and P strength at 0.9 and 1.4 V; tempco falls with N strength)

178.0 - 279.2 MHz
f at 0.9 V across the 9 textbook corners

306 - 1151
textbook tempco range ppm/C (3.8x) vs PDK 5-corner 112 - 1399 (12.5x)

Takeaway: with N and P varied independently the ordering is textbook-clean; only the PDK skews break the tempco range.

Audit 4: Tempco Follows NMOS Drive



-0.78 / -1.1
Vth tempco mV/C, NMOS / PMOS (textbook -0.5 to -3)

-30 % / +14 %
NMOS Id over 125 C at 1.8 V / 0.9 V

-816 ppm/C
tt tempco at V_ctrl = 1.4 V (sign flips from +681 at 0.9 V)

Takeaway: weaker NMOS drive means larger positive tempco; the bias sits below the zero-tempco point.

Tempco Mechanism and Physics Checks

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Why the frequency rises with temperature at 0.9 V

Vth falls with temperature (about -0.8 mV/C NMOS). With only ~0.2 V overdrive, that gain wins over the mobility loss, so drive current rises.

Which device limits the ring depends on V_ctrl

At 0.9 V the NMOS dominates (36 % spread slow to fast N vs 10 % across P). At 1.4 V the PMOS dominates (26 % vs 9 %).

Other checks that pass

Supply current is proportional to f in every corner (C ~ 27-30 fF per node). Closed-loop lock voltage equals the open-loop value for 226.2 MHz within 0.2 mV at all 11 corners.

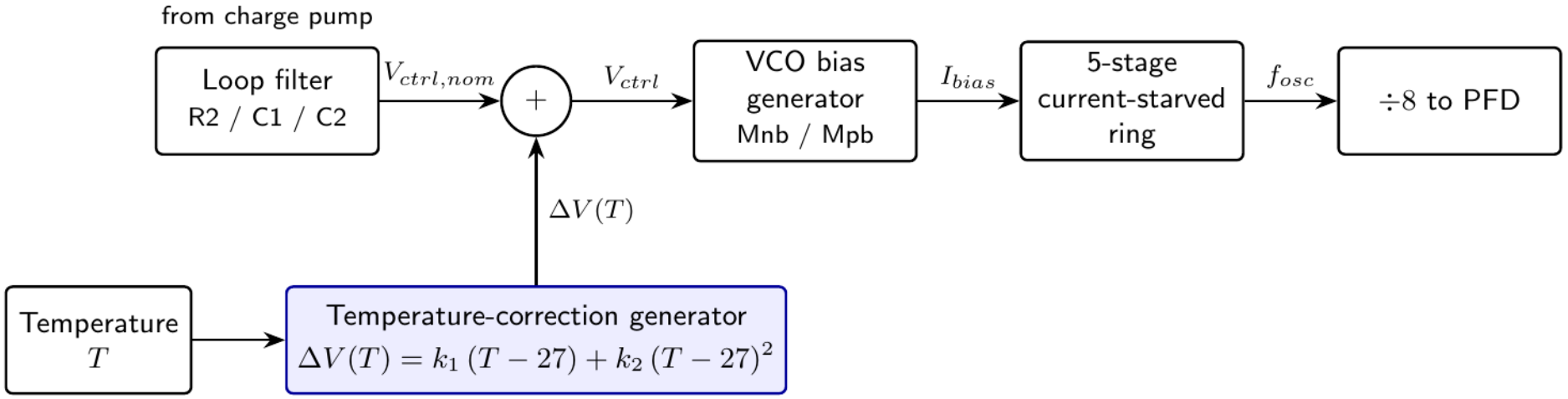
36 % vs 10 %
0.9 V: NMOS vs PMOS spread

26 % vs 9 %
1.4 V: PMOS vs NMOS spread

0.2 mV
lock V vs open-loop V, 11 corners

Takeaway: the tempco mechanism is overdrive, not N/P balance; every physics check agrees with theory.

Compensation Architecture



1st-order: $k_2 = 0$ (linear, PTAT/CTAT-equivalent) **2nd-order:** $k_2 \neq 0$ (curvature-matched)
Coefficients fitted per process corner; an ideal behavioural source in Review 2.

dV(T)
correction added to V_ctrl before the VCO bias generator

1st order
k1 (T - 27): PTAT/CTAT equivalent

2nd order
adds k2 (T - 27)^2 for the curvature

Ideal
behavioural source; transistor-level network is future work

Takeaway: the correction is a voltage added to V_ctrl: v_eff = v + k1 (T-27) + k2 (T-27)^2.

Compensation Method: Fit, Then Verify

scripts/comp_factorial.py, applied per corner

1. Measure f(T, V) at V = 0.85 / 0.9 / 0.95 V for 6 temperatures.
2. Solve the exact voltage shift dV*(T) that restores f(27 C).
3. Least-squares fit k1 (1st order) and k1, k2 (2nd order).
4. Verify by simulating the compensated VCO at the 6 fit temperatures and at 4 held-out temperatures (-30, 10, 40, 70 C).

Why Review 2's numbers changed

Review 2 used a wrong K_VCO (669 instead of ~890 MHz/V) and an analytic first-order guess. The measured fit replaces it: tt span 1.12 % → 0.20 %.

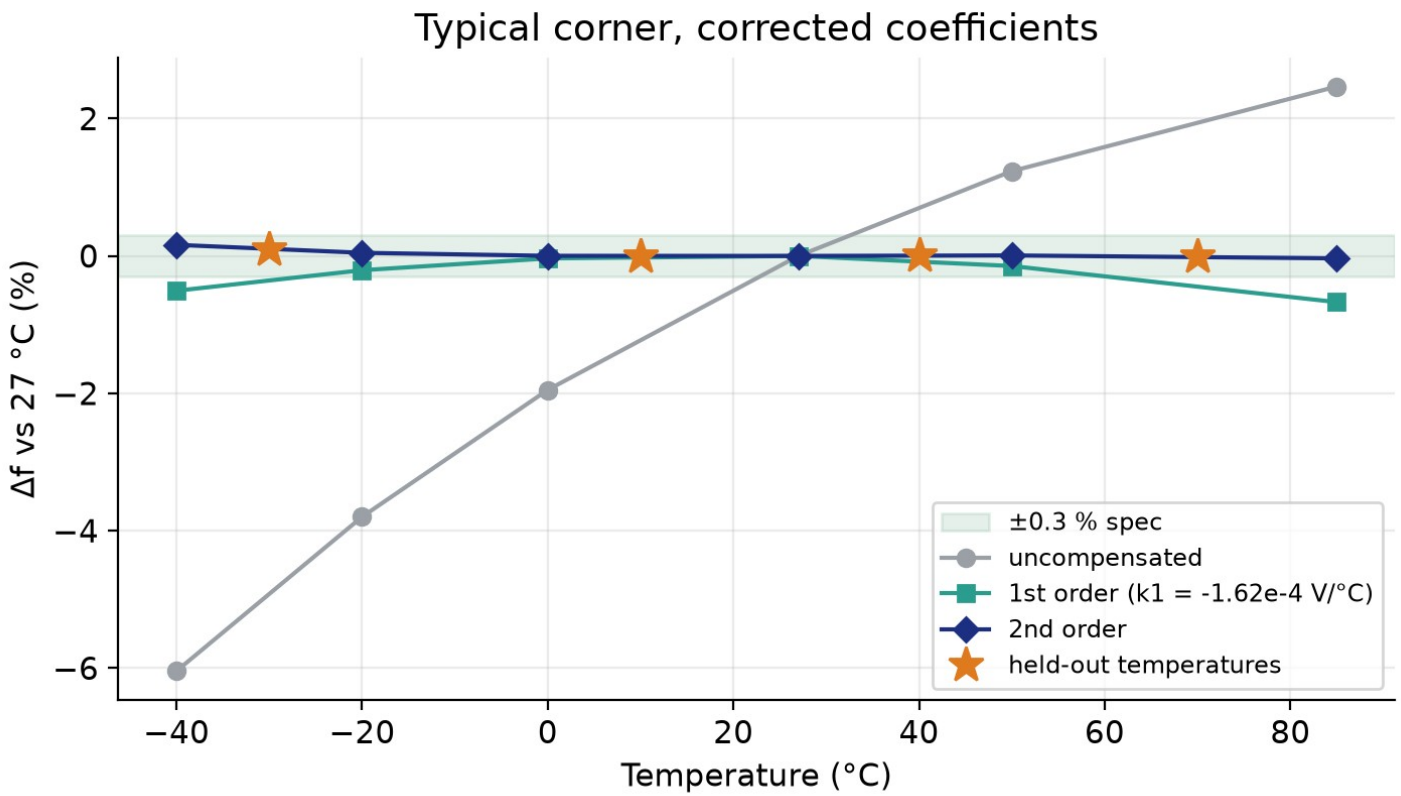
-1.543e-4
k1 at tt (V/C)

4.18e-7
k2 at tt (V/C^2)

4 held-out
temperatures never used in the fit

Takeaway: coefficients come from measured data and are checked at temperatures the fit never saw.

Compensation at the Typical (tt) Corner



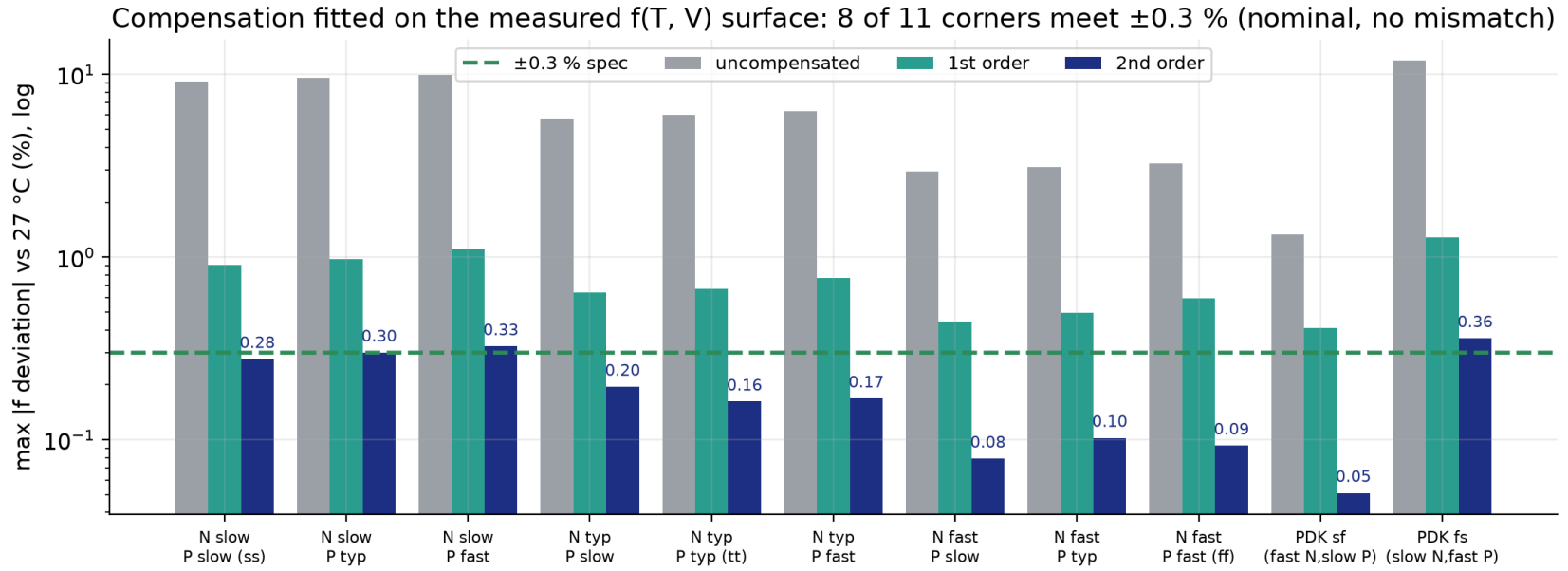
8.51 → 0.20 %
span: baseline → 2nd order (1st order 0.67 %)

0.16 %
max deviation; +/-0.3 % spec MET

0.09 %
max deviation at held-out temperatures

Takeaway: at tt the compensated VCO stays within +/-0.16 % over -40 to 85 C.

Compensation on 11 Corners



8 / 11
corners meet +/-0.3 % (7 of 9 textbook)

31 - 44x
compensation ratio (PDK sf 15.9x)

0.03 - 0.25 %
held-out max deviation

Takeaway: 8 of 11 corners meet the spec; the slow-NMOS corners are the hardest.

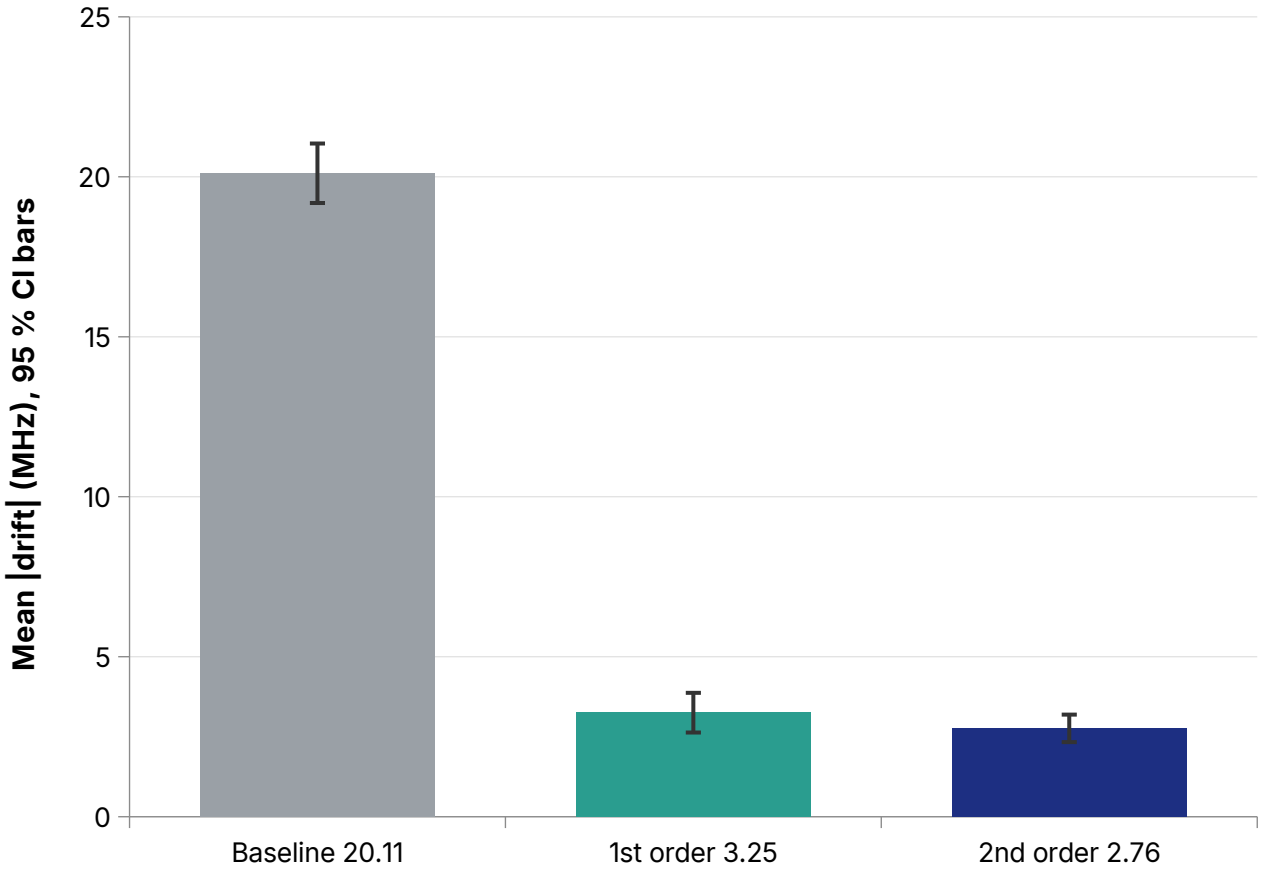
Compensation Results: All 11 Corners

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Corner	f 0.9 V (MHz)	Base span %	1st %	2nd %	Max dev %	Held-out %	Ratio C	+/-0.3 %
N slow / P slow (ss)	178.0	13.37	0.91	0.35	0.277	0.175	38.4	yes
N slow / P typ	188.0	13.97	0.98	0.38	0.301	0.201	37.2	NO
N slow / P fast	195.9	14.38	1.10	0.42	0.325	0.223	34.3	NO
N typ / P slow	211.8	8.10	0.64	0.24	0.195	0.109	33.6	yes
N typ / P typ (tt)	226.2	8.51	0.67	0.20	0.162	0.091	43.2	yes
N typ / P fast	238.1	8.79	0.77	0.20	0.168	0.097	43.7	yes
N fast / P slow	242.8	3.82	0.45	0.12	0.079	0.034	31.0	yes
N fast / P typ	262.7	4.06	0.50	0.12	0.102	0.041	33.3	yes
N fast / P fast (ff)	279.2	4.15	0.59	0.11	0.093	0.035	37.0	yes
PDK sf (N fast / P slow)	246.2	1.49	0.41	0.09	0.051	0.036	15.9	yes
PDK fs (N slow / P fast)	183.7	17.49	1.29	0.49	0.360	0.251	35.8	NO

Takeaway: NOT met: N slow / P typ 0.301 (borderline), N slow / P fast 0.325, PDK fs 0.360. Nominal, ideal source.

Monte Carlo (n = 72 chips, tt): Mean |Drift|



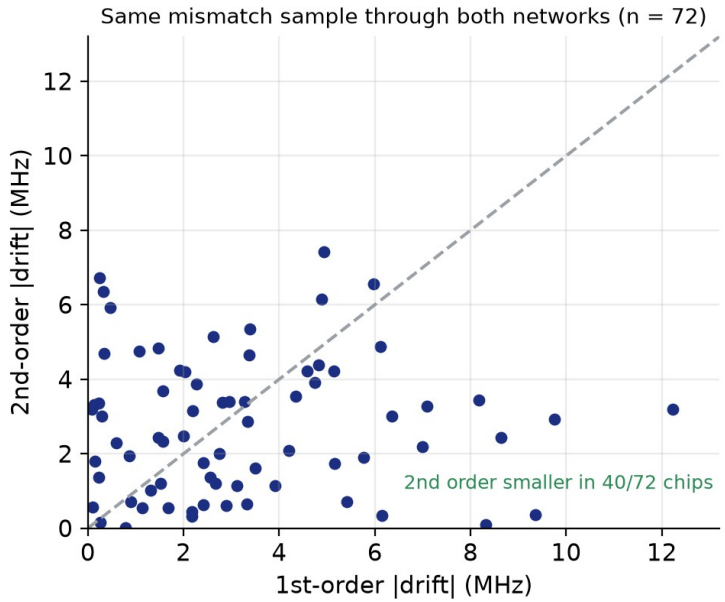
-17.35 MHz
baseline → 2nd order, CI [16.37, 18.34], 72/72 chips

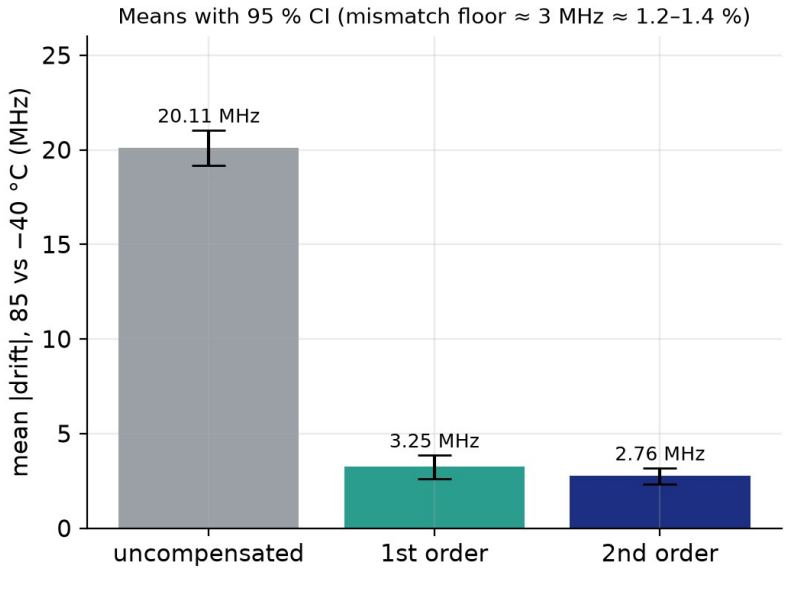
-16.86 MHz
baseline → 1st order, CI [15.57, 18.16], 71/72 chips

0.49 MHz
1st vs 2nd order: CI [-0.26, 1.24], NOT significant (2nd smaller in 40/72)

Takeaway: compensation cuts drift ~7x (72/72 chips), but the 2nd-order term adds no detectable benefit under mismatch.

Monte Carlo: The Mismatch Floor





~3 MHz
residual drift (~1.2-1.4 % of 226 MHz)

0.2 - 0.7 %
nominal residual without mismatch

Per-chip trim
needed to reach +/-0.3 % across chips (future work)

Takeaway: transistor mismatch makes each chip drift differently, and one global correction cannot remove it.

Loop Dynamics: Found, Fixed, Verified

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Found

The Review-1 loop design dropped a 1/(2 pi) factor in omega_n^2, so damping was zeta ~0.33, not the 0.72 claimed. The original loop had not settled at ss/ff/sf/fs within 2 us.

Fixed

R2 retuned 1.7 k → 3.6 kOhm: zeta ~0.70, phase margin ~65-70 deg (100 x zeta rule, not a Bode plot).

Cost

Cold start is slower (tt 839 → 1351 ns) but in spec; coarse start faster (577 → 503 ns).

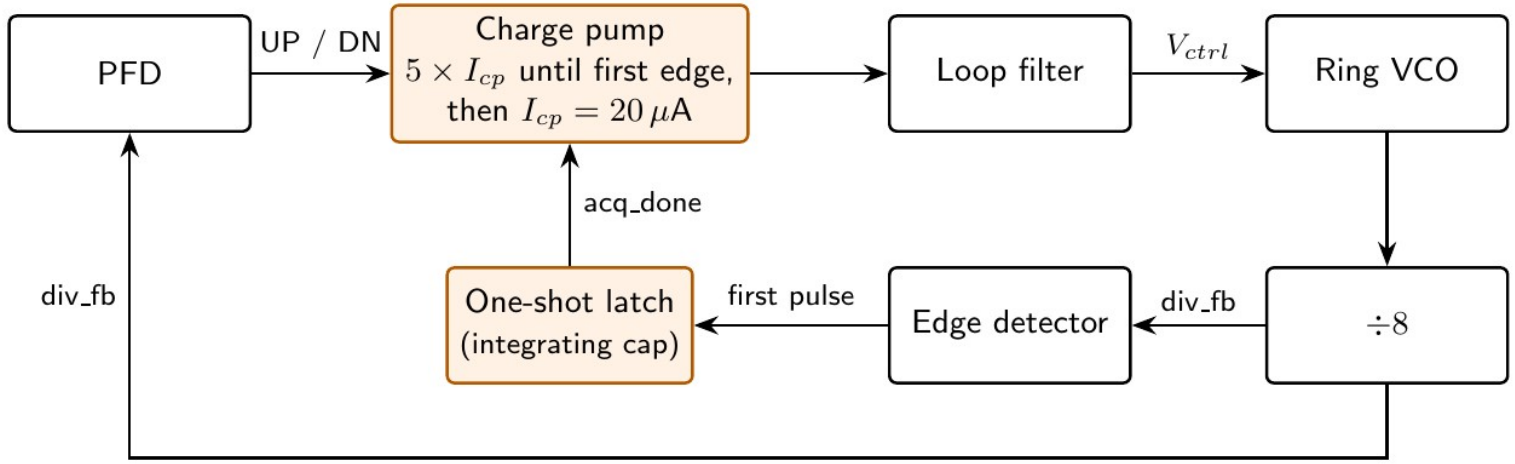
zeta 0.33 → 0.70
damping

8.6 → 0.35 mV
V_ctrl ripple at lock

0.78 → 0.008 %
final frequency error

Takeaway: we caught a design-equation error ourselves; the retuned loop is well damped and quiet.

Cold-Start Fix: Boosted Charge Pump



Problem

From 0 V the VCO is in deep subthreshold (0.6 MHz at 0.5 V, 6.5 MHz at 0.6 V vs 226 MHz target), so the divider clock is effectively absent and the pump-up is slow.

Fix

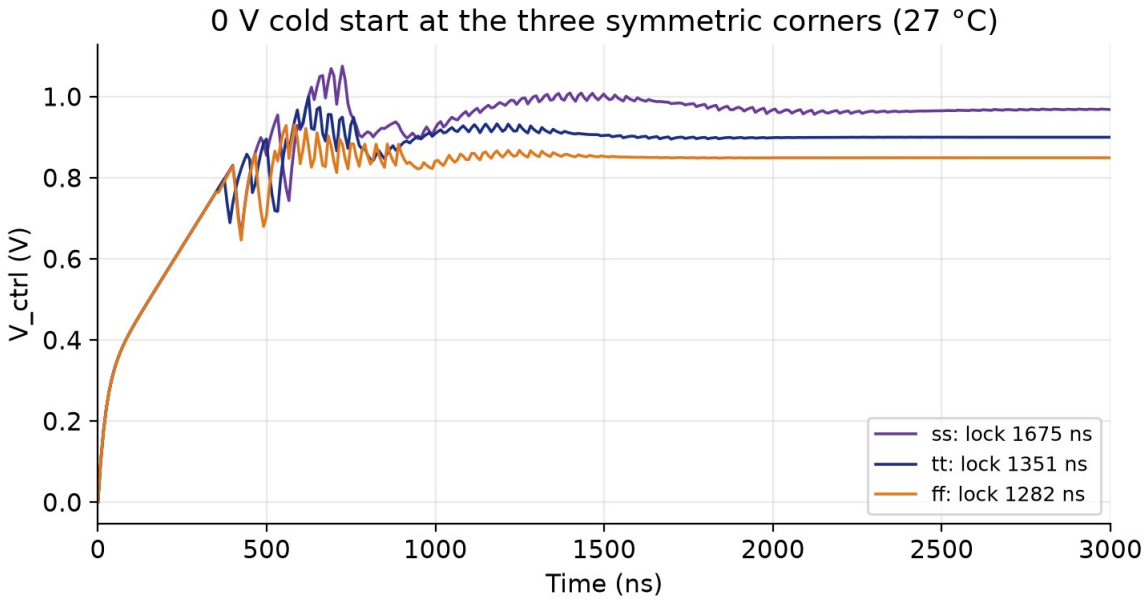
Charge pump runs at 5 x I_cp until the divider's first edge, then drops to 20 uA.

1351 ns
tt lock from 0 V

503 ns
tt lock from 0.85 V

Takeaway: the boosted pump removes the dead zone at start-up; lock is inside 2 us.

Cold-Start Traces



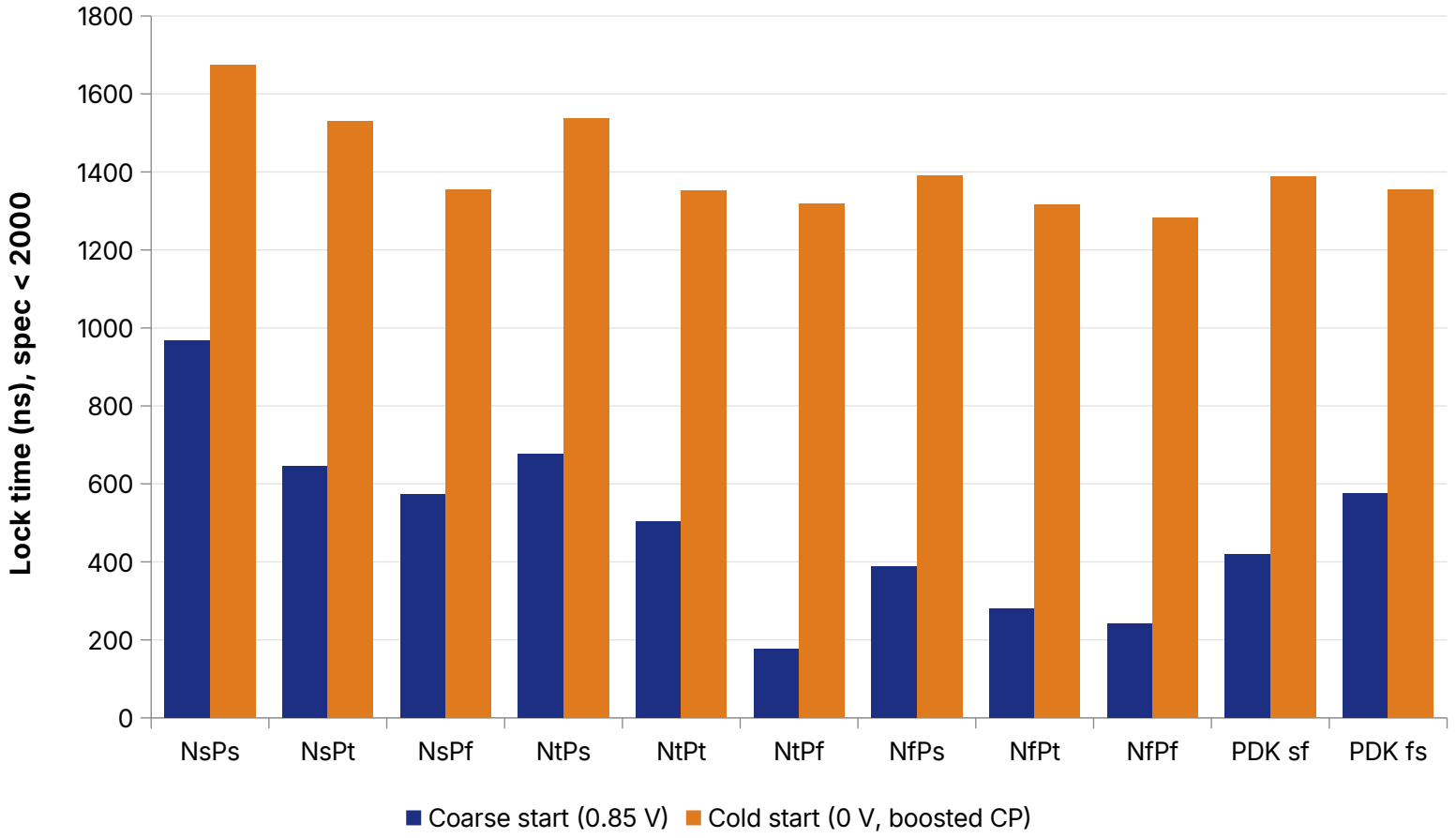
1283 - 1675 ns
cold-start lock, 11 corners, 27 C

<= 0.8 mV
V_ctrl ripple in every run

226.17 MHz
= 8 x f_ref (-0.01 %) in every run

Takeaway: from 0 V the control voltage climbs and settles to its corner-specific value.

Lock Time Across 11 Corners (27 C)



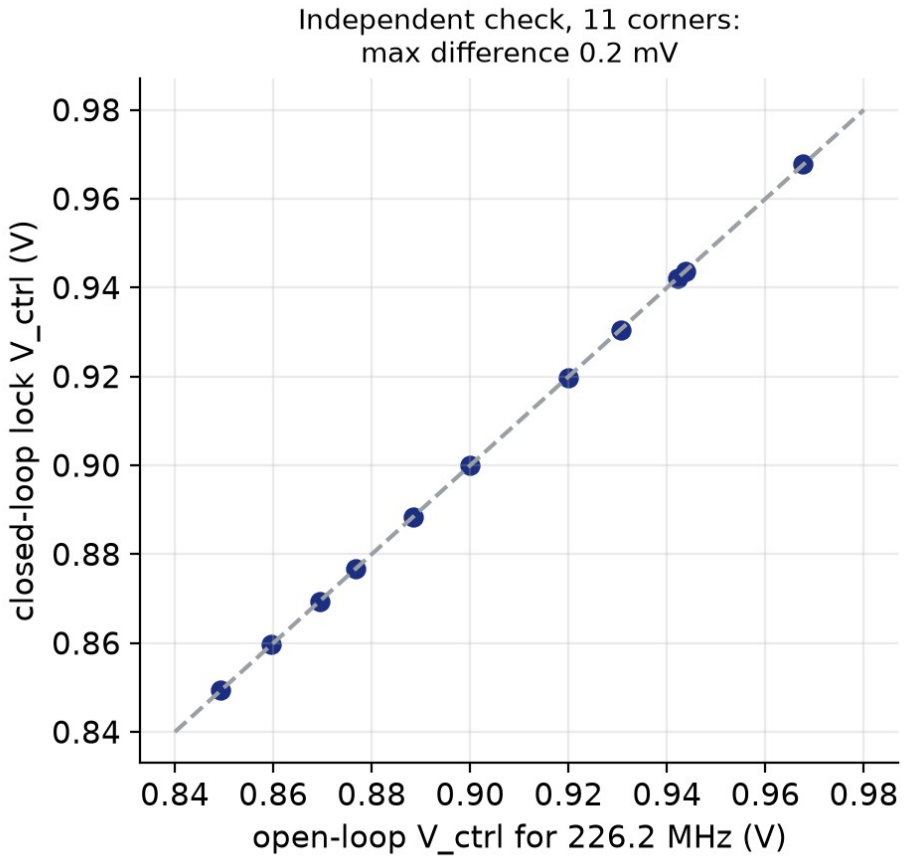
177 - 967 ns
coarse start

1283 - 1675 ns
cold start

22 / 22
runs inside 2 us at 27 C

Takeaway: all 11 corners lock inside 2 us at 27 C; slow-NMOS corners take longest. (Ns = N slow, Pf = P fast, etc.)

Does the Loop Lock at the Right Voltage?



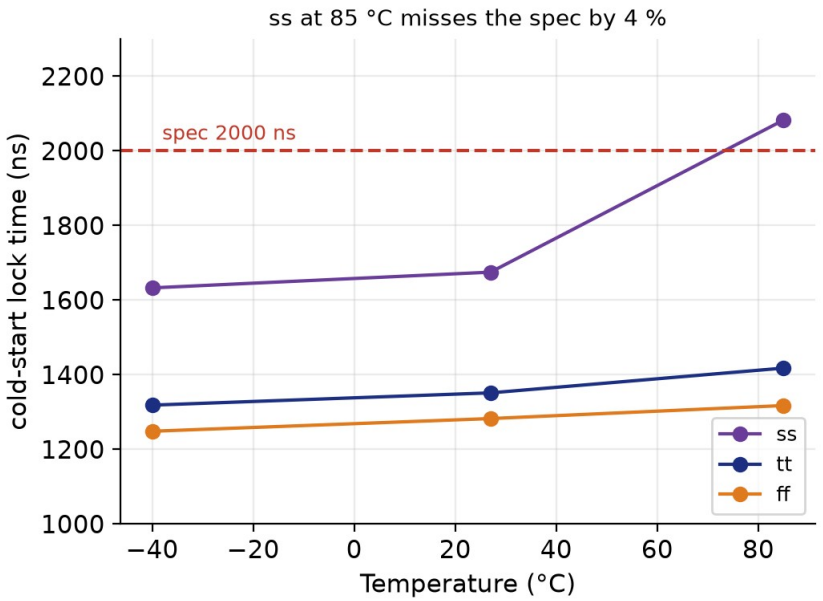
< 0.2 mV
lock V vs open-loop V for 226.2 MHz, 11 corners

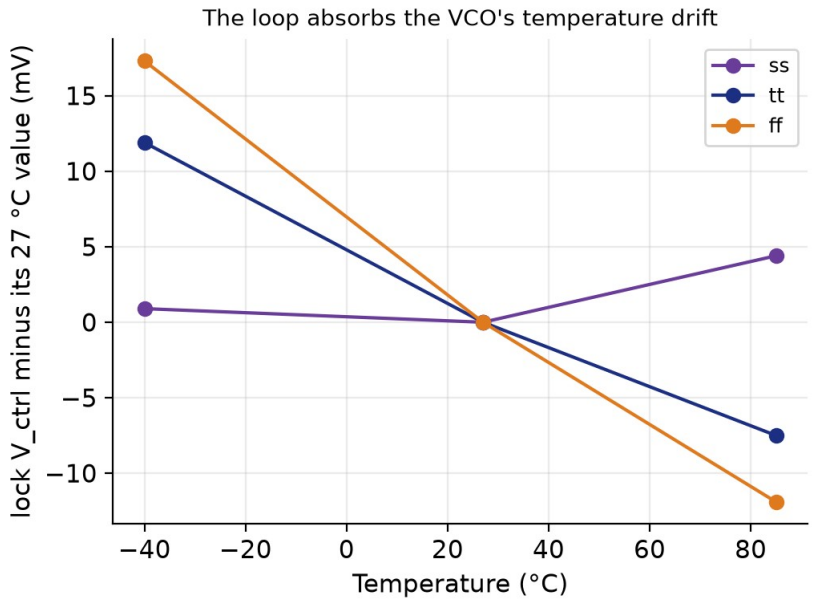
0.849 - 0.968 V
lock V_ctrl range, ff to ss at 27 C

+11.9 / -7.5 mV
tt lock-V shift at -40 / 85 C = fitted compensation (+12.2 / -7.5)

Takeaway: the PLL absorbs the VCO tempco; at tt this independently confirms the compensation fit.

Temperature: Cold-Start Lock at -40, 27, 85 C





2083 ns
ss at 85 C: MISSES the 2 us spec by 4 %

1319 / 1418 ns
tt at -40 / 85 C

1249 / 1317 ns
ff at -40 / 85 C

Takeaway: worst case ss / 85 C misses the lock spec slightly; every other case is inside it.

Benchmark vs Literature (Honest)

Design	Process / basis	Residual drift	Lock time
This work, 2nd order (tt, nominal)	sky130, simulation, ideal source	+/-0.16 % max dev (0.20 % span), -40 to 85 C	1351 ns (0 V cold start)
This work, 2nd order (MC n = 72)	sky130, simulation, mismatch on	~2.76 MHz = ~1.2 % |drift|, 85 vs -40 C	n/a
Souliotis 2022 [1]	180 nm, measured silicon	-1.5 % / +0.4 %, -40 to 120 C	not reported
Stancu 2023 [2]	simulation, 50 MHz clock	1.6 % temperature error	n/a
Geng 2018 [3]	180 nm, measured silicon	not reported as drift	< 2 us

Nominal simulation of an ideal network is competitive, but it is not like-for-like with measured silicon or with a transistor-level circuit. With chip-to-chip mismatch our residual (~1.2 %) is the same order as the published numbers.

No jitter or phase-noise number yet (no PSS/Pnoise in open tools), so [3]'s -117 dBc/Hz cannot be compared.

Takeaway: in the right ballpark, not proof of parity with fabricated designs.

What We Found and Corrected

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#	Earlier claim	Corrected
1	sf / fs labels (sf = slow N)	reversed: PDK sf = fast N / slow P (measured)
2	Tempco set by N/P balance	set by NMOS overdrive; sign flips at 1.4 V
3	K_VCO = 669 MHz/V	~890 MHz/V (887 local slope at 0.9 V)
4	Hard oscillation floor at 0.63 V	none: 0.63 MHz at 0.5 V, 6.5 MHz at 0.6 V (window artefact)
5	Loop zeta 0.72 (1/(2 pi) dropped)	zeta 0.33 → retuned R2 = 3.6 k gives ~0.70
6	Compensation coefficients (wrong K_VCO)	refit: tt 0.16 % max dev, 8 / 11 corners meet spec
7	2nd order significantly better (n = 30)	n = 72, corrected fit: not significant; mismatch floor ~1.2-1.4 %
8	PDK skew corners lie inside ss-ff	wider than ss/ff; textbook 3 x 3 set added
9	Closed loop fine in all cases	ss / 85 C cold start 2083 ns misses 2 us by 4 %

Takeaway: each fix came from our own audit and each is traceable to a re-run.

Known Limitations

Compensation network and charge pump are behavioural (ideal) models; no transistor-level versions yet.

No layout, DRC, LVS or post-layout simulation yet.

No jitter or phase-noise number (no PSS/Pnoise in open tools).

Phase margin is the 100 x zeta rule, not a Bode plot.

Only R2 = 1.7 / 3.0 / 3.6 kOhm simulated at transistor level; 3.6 k is not claimed optimal.

Cold start simulated on 11 corners at 27 C and at 3 corners for +/-temperature, not all 11 x all T.

Compensation is fitted nominally at V_ctrl = 0.9 V; each corner actually locks at 0.85 - 0.97 V.

Mismatch limits compensated drift to ~1.2-1.4 %; the H1 test and Spectre arm are pending.

Takeaway: we state plainly what is ideal, nominal, or still missing.

Remaining Work / Next Phase

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Circuit

Transistor-level charge pump and compensation network (PTAT/CTAT + squarer), then re-run the loop.

Define the compensation at each corner's own lock voltage; per-chip trim to beat the mismatch floor.

Physical design

Layout in Magic, DRC, LVS (netgen, built), post-layout simulation.

Validation and release

Virtuoso/Spectre cross-check arm, PSS/Pnoise, H1 regression (needs department Virtuoso access - to confirm with Dr. Bindu).

Spec sign-offs (V_ctrl floor revision, +/-0.15 ppm restatement); paper and artifact release.

Takeaway: the behavioural result is solid; the next phase makes it real hardware-shaped.

Final Review at a Glance

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8 / 11
corners meet +/-0.3 % (tt 0.16 % max dev)

8.51 → 0.20 %
tt span, baseline → 2nd order

31 - 44x
compensation ratio (PDK sf 15.9x)

20.1 → 2.8 MHz
Monte Carlo |drift|, n = 72

1.2 - 1.4 %
mismatch floor (per-chip trim needed)

177 - 967 ns
coarse-start lock, 11 corners, 27 C

1283 - 1675 ns
cold-start lock, 11 corners, 27 C

2083 ns
ss / 85 C cold start: misses 2 us by 4 %

zeta ~0.70 (R2 = 3.6 k); 15 references; 1 patent narrowed the claim; 9 corrections found by our own audit.

Takeaway: a verified, honest closed-loop study with its limits stated.

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Thank You

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